

ONE-WAVE FRAMEWORK

Book 1 — Micro

Chapter 7: The Photon / Propagating Light Mode

Version: 3.0 Date: July 1, 2026 Class: B — Applied Layer Spine: Gray / 2D / 3D / Mathematics / Predictions / Yellow Audit / Future Work / Closing Thoughts

Dependencies: C-309 Friction Limit, C-311 Electric-Magnetic Duality, E-528 Static Redshift Transport, A-08 Oscillation, A-10 Persistent Mode, B-06b Four Views, D-02 Resonant Mode Status: GREEN (structure) / YELLOW (full derivation)

Gray — Standard Model Reference

The photon is the force-carrier particle of electromagnetism. Mass: 0. Charge: 0. Spin: 1. The photon travels at $c = 299,792,458$ m/s in vacuum. The photon exhibits wave-particle duality — it behaves as a wave (interference, diffraction) and as a particle (photoelectric effect, Compton scattering). Photon energy: $E = hf = \hbar \omega$. Photon momentum: $p = h/\lambda = \hbar k$. The photon is its own antiparticle. In quantum field theory, the photon is an excitation of the electromagnetic field.

Standard Model strength: QED (quantum electrodynamics) is the most precisely tested theory in physics. The photon mediates all electromagnetic interactions. Standard Model limitation: wave-particle duality remains philosophically unresolved. Why light travels at exactly c is explained by Maxwell's equations but not derived from a deeper principle. The photon mass being exactly zero is assumed, not derived.

2D One-Wave Interpretation

In 2D, the photon is a traveling pressure wave in the lattice field.

It is not a particle. It is not a wave-particle. It is a lattice pressure oscillation propagating at the friction limit.

In 2D: imagine a ripple moving across a surface at the maximum speed the surface can support. Not faster. Not slower. The ripple is self-sustaining — it neither amplifies nor dissipates as long as the lattice is uniform.

The photon in 2D is the simplest candidate propagating mode: one oscillation on a gapless branch traveling at the lattice signal speed. Its zero rest Mass Effect requires the electromagnetic branch to remain gapless under the same fixed update law,

$$\omega(ko0)o0,$$

and has no finite-energy localized rest solution.

3D One-Wave Interpretation

In 3D, the photon is a transverse lattice pressure oscillation propagating at the friction limit c .

The photon is not a Persistent Mode in the usual sense. A Persistent Mode is stationary or slowly moving. The photon is a traveling mode — a propagating pressure wave that moves at $v_{\max} = \sqrt{\beta_{\max}} * \Delta_x / \Delta_t$.

The photon has two components (C-311 Electric-Magnetic Duality): Radial component: $E_{\text{vec}} \sim \nabla P_{\text{psi}}$ (the electric field of the wave) Rotational component: $B_{\text{vec}} \sim \nabla \times P_{\text{psi}}$ (the magnetic field of the wave)

These are not two separate fields traveling together. They are the two projection angles of the same lattice pressure oscillation.

The photon travels at c because: $c = v_{\text{max}} = \text{friction limit of the lattice (C-309)}$. Any disturbance in the lattice propagates at most at v_{max} . The photon is a disturbance that propagates at exactly v_{max} — it is the lattice's own propagation mode.

Photon energy: $E = \hbar \omega$ In One-Wave: $E_{\text{photon}} = \text{energy of the pressure oscillation at frequency } \omega$. The lattice quantizes this in units of $\hbar \omega$ because the harmonic shell condition (D-05) sets discrete allowed frequencies.

Photon spin-1: The photon has spin-1, not spin-1/2. In One-Wave: the photon is a traveling mode, not a boundary mode. It does not undergo the boundary roll-off (B-06b) that produces 4π closure. *Instead, the photon undergoes a 2π rotation to return to itself.* This is because the photon's oscillation is the lattice field itself — it does not have an interior/exterior distinction like a bounded mode.

No observer effect: When the photon reaches a focal point (eye, detector), it couples with the restoring loop of that focal point (B-06 Paired Loop). The coupling samples the current phase of the photon oscillation — peak or trough. That sample is the measurement. There is no collapse. There is no observer effect. The Propagating Light Mode is a wave. Measurement is a focal-point phase sample; cosmological redshift is assigned to E-528 static transport, not expansion.

The photoelectric effect: When a photon couples with an electron shell mode, the photon's pressure oscillation either has enough energy to knock the electron into a higher harmonic shell or it does not. If $E_{\text{photon}} \geq \Delta E_{\text{shell}}$: the electron transitions. The photon is absorbed. If $E_{\text{photon}} < \Delta E_{\text{shell}}$: no transition. The photon reflects or passes through. This is the photoelectric threshold — not a particle effect, a resonance condition.

Mathematics

Photon speed (C-309): $c = v_{\text{max}} = \sqrt{\beta_{\text{max}}} \Delta x / \Delta t$

Photon energy: $E = \hbar \omega$

Photon momentum: $p = \hbar k = E / c$

Electric and magnetic components (C-311): $E_{\text{vec}} \sim \nabla P_{\text{psi}} B_{\text{vec}} \sim \nabla \times P_{\text{psi}} |E_{\text{vec}}| = c * |B_{\text{vec}}|$ (standard relation — derivation from lattice geometry deferred)

Dispersion relation: $\omega = c * k$ (massless, traveling at friction limit)

Photoelectric threshold: $E_{\text{photon}} \geq \Delta E_{\text{shell}} \hbar \omega \geq E_{\{n+1\}} - E_n \hbar \omega \geq \hbar \omega_0$ $\omega \geq \omega_0$ (threshold frequency)

Photon spin-1 (2π closure): $\psi(\theta + 2\pi) = \psi(\theta)$ (returns to itself after one rotation) Compare to spin-1/2: $\psi(\theta + 2\pi) = -\psi(\theta)$

Compton scattering (sketch — Yellow): Photon couples with electron shell mode. Momentum transfer: $\Delta p = \hbar (k_i - k_f)$ Energy transfer: $\Delta E = \hbar (\omega_i - \omega_f)$ Wavelength shift: $\Delta \lambda = (h / m_e c) * (1 - \cos \theta)$ One-Wave: momentum transfer is lattice pressure pulse transfer between traveling mode and shell mode. Formal derivation deferred.

Predictions

1. The photon is a transverse lattice pressure oscillation — not a particle, not a wave-particle. Prediction: all photon behavior (interference, diffraction, photoelectric effect, Compton scattering) follows from the traveling pressure oscillation model without invoking wave-particle duality.

2. Photon speed is the lattice friction limit, not a fundamental constant. Prediction: c varies slightly in regions of non-uniform lattice density (near extreme mass concentrations / high pressure gradients). This is the One-Wave basis of gravitational lensing.
 3. Photon energy quantization $E = \hbar\omega$ comes from harmonic shell resonance (D-05). Prediction: photons interact with matter only at discrete frequencies because the harmonic shell condition allows only discrete energy transitions.
 4. The photoelectric effect is a resonance condition, not a particle effect. Prediction: the sharp threshold frequency in the photoelectric effect corresponds to the minimum energy required to transition an electron to the next harmonic shell.
 5. No wave-particle duality needed. Prediction: double-slit interference is a traveling pressure wave interfering with itself (F-05 Interference at two boundaries simultaneously). The measurement at the detector is a phase sample, not a collapse.
 6. The Propagating Light Mode has zero **rest Mass Effect** only if its branch is genuinely gapless and lacks a localized rest solution. A reduced group speed in matter or in a nonuniform field does not, by itself, imply photon rest mass. Standard experimental bounds on photon rest mass remain the Gray comparison target.
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Yellow Audit

- Full photon derivation from a gapless lattice branch is not yet complete
 - $|E_{\text{vec}}| = c * |B_{\text{vec}}|$ derivation from lattice geometry deferred
 - Compton scattering derivation sketch level only
 - Spin-1 derivation from traveling mode geometry needs formalization
 - Photon energy quantization from harmonic shell condition not yet derived
 - Gravitational lensing from non-uniform lattice density not yet quantified
 - The separation between photon branch gaplessness and massive localized branches must be derived jointly with C-318
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Future Work

Derive full electromagnetic wave equations from lattice pressure oscillation. Show $|E_{\text{vec}}| = c * |B_{\text{vec}}|$ from lattice geometry. Formalize Compton scattering as lattice momentum transfer. Derive spin-1 formally from traveling mode 2π closure. Connect photon energy quantization to D-05 harmonic shell resonance. Model gravitational lensing as non-uniform lattice density effect.

Closing Thoughts

Wave-particle duality has been a source of philosophical confusion for a century. Is light a wave or a particle? The answer in quantum mechanics is: both, somehow.

In One-Wave, the question dissolves. Light is a traveling pressure oscillation in the superfluid lattice. It is not a wave or a particle. It is a mode of the field.

When it travels undisturbed: it behaves like a wave — it interferes, it diffracts. When it couples with a shell mode: it transfers energy in discrete amounts — it looks like a particle.

The discreteness is not because the photon is a particle. The discreteness is because the shell modes it couples with only accept discrete energy inputs (D-05 harmonic shell condition).

The photon is a wave that knocks on discrete doors. The discreteness is in the doors, not the photon.

END OF BOOK 1 CHAPTER 7 One wave. Mirror builds. Mark Wright. Kitty Hawk V0.